

Deterministic pose-spline dynamics estimator

Shared parameter update: nominal -> estimated

mass [kg]	2.351597591 -> 2.770070676	delta=+0.418473
ratio=1.1779527		
CoG x [m]	0 -> -0.002197569206	delta=-0.00219757 ratio=-
CoG y [m]	0 -> 0.0007344321885	delta=+0.000734432 ratio=-
CoG z [m]	0 -> -0.02207523269	delta=-0.0220752 ratio=-
rotor effectiveness 1	1 -> 0.9918543383	delta=-0.00814566 ratio=0.99185434
rotor effectiveness 2	1 -> 1.044524479	delta=+0.0445245 ratio=1.0445245
rotor effectiveness 3	1 -> 0.9785668916	delta=-0.0214331 ratio=0.97856689
rotor effectiveness 4	1 -> 0.9863770579	delta=-0.0136229 ratio=0.98637706
command lag [s]	0.01 -> 0.03886824096	delta=+0.0288682 ratio=-

Inertia matrix [kg m²]

nominal:
[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]
estimated:
[0.07236800026, -0.002152126932, 6.605126651e-05]
[-0.002152126932, 0.1541579028, 0.0001600046787]
[6.605126651e-05, 0.0001600046787, 0.2255612778]

Principal moments: nominal -> estimated

principal moment 1 [kg m ²]	0.06495264516 -> 0.0723113786	delta=+0.00735873
ratio=1.1132938		
principal moment 2 [kg m ²]	0.06500006701 -> 0.1542141414	delta=+0.0892141
ratio=2.3725228		
principal moment 3 [kg m ²]	0.1289921163 -> 0.2255616608	delta=+0.0965695
ratio=1.7486469		

Optimization

joint dynamics loss: 0.178477846339
soft-prior cost: 0.0531116909237
selected strict-ZOH lag [s]: 0.038868240958

Bag failure_1

reconstruction position RMSE [m]: 0.000511441948809
reconstruction position component RMSE [m]: [0.000264 0.000287 0.000331]
reconstruction orientation RMSE [deg]: 0.491033445648
gyro consistency RMSE xyz [rad/s]: [0.050385 0.085584 0.038392]
specific-force consistency RMSE xyz [m/s²]: [0.956581 0.162707 0.351708]
inferred external force RMS xyz [N]: [0.560197 0.317037 1.210018]
inferred external torque RMS xyz [N m]: [0.070902 0.49178 0.14326]
bag dynamics loss: 0.274763331312

Bag failure_2

reconstruction position RMSE [m]: 0.000368116986919
reconstruction position component RMSE [m]: [0.000186 0.000244 0.000204]
reconstruction orientation RMSE [deg]: 0.245016993583
gyro consistency RMSE xyz [rad/s]: [0.028058 0.036338 0.033608]
specific-force consistency RMSE xyz [m/s²]: [0.342308 0.097953 0.086711]
inferred external force RMS xyz [N]: [0.187281 0.384931 0.670126]
inferred external torque RMS xyz [N m]: [0.056473 0.229705 0.085723]

bag dynamics loss: 0.082192361367